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Friendshoring and Trade: Analysing Trade Flow Shifts in the Post-Trump Era and the Relevance of Geopolitics in International Trade

Final Project Report

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Abstract

Recent geopolitical tensions have spurred debate over whether trade patterns increasingly reflect political alignment rather than economic efficiency—a phenomenon termed "friendshoring." This project investigates whether geopolitical alliances have evolved into comprehensive strategic partnerships that reshape international trade flows. Using dyadic data on political, cultural, religious, linguistic, and geographic distances alongside IMF trade statistics from 2015 and 2021, we apply principal component analysis and hierarchical clustering (Ward linkage) to group countries based on numerous such features. We complement this with a first-difference regression in an attempt to quantify the effect of geopolitical and cultural shifts on trade in particular. Our clustering results reveal stable blocs that persist across both years, yet fail to correctly identify contemporary geopolitical rivals. For example, grouping Russia with NATO members suggests that trade linkages may be more resilient to political discord than commonly thought. The regression shows weak explanatory power, highlighting the limited role of political distance in trade changes relative to other factors. Methodologically, this work demonstrates how machine-learning techniques can be integrated with traditional econometric tools to explore complex questions in international political economy, providing a reproducible pipeline for future interdisciplinary research.

Keywords: data science, Python, machine learning, hierarchical clustering, Ward linkage, friendshoring, trade

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1 Introduction

Standard trade theory has long established that it is necessity, endowments, and comparative advantage that should drive the flow of goods between nations. Taking nations to be rational actors, such a theory suggests governments are wont to exercise trade policy in a manner that serves free trade ideals. This approach has guided international trade mechanisms for decades now, with free trade being enshrined by organisations like the World Trade Organisation (WTO). Theories in favour of protectionism, such as mercantilism, are often heralded as outdated and medieval.

However, recent events indicate a turn away from such free trade ideals, with the disbandment of the WTO's Appellate Body and the increasing practice of using trade flows as a geopolitical bargaining tool. This is seen by the broad-spectrum tariffs introduced by the United States (US), with subsequent reductions in tariff rates being used as rewards for nations that comply, or the leveraging of rare earth metals exports by China against nations perceived as unfriendly.

In this global environment, "friendshoring" has become a buzzword in trade discussions, referring to the phenomenon of global trade flows being dictated increasingly by geopolitical considerations rather than economic pragmatism, opposing the traditional views in favour of free trade and globalisation. A study conducted by the McKinsey Global Institute finds early evidence of friendshoring, pointing to the United States' reduction of the "geopolitical distance" covered by its trade by 10% since 2017 — just one year after the election of Donald Trump [1]. The use of trade policy as a geopolitical tool has since become increasingly common, exacerbated by the U.S.–China Rivalry, the Russo–Ukrainian War, the rush to secure supply chains after the COVID–19 Pandemic, and numerous other such frictions.

This project aims to investigate this phenomenon through a machine learning approach, asking the following question: Have geopolitical alliances evolved into comprehensive strategic partnerships — with alignment spilling over into non–geopolitical realms — since the election of President Trump? And to what extent has this multidimensional alignment reshaped global trade flows? The paper proceeds by firstly conducting a brief literature review on related work, providing theoretical background on friendshoring and motivating the use of machine learning on such areas. Then, it will provide a description of the data and the methodology of our technical approach. Finally, the paper presents the results of our analysis before concluding.

2 Literature Review

The rise of friendshoring is often seen as a part of a broader retreat from globalisation and a turn towards tribalism, which refers to the rise in nationalist or populist movements that often result in "us-vs-them" identities. Such patterns have been observed increasingly in recent years across nations as diverse as the United States, France, Italy, Hungary, Poland, Brazil, India, and the Philippines, among others. While there is a tendency to attribute such changes to "right-wing politics", the national studies conducted by the World Economic Forum suggest a different story — one stemming from deepening mistrust in society and institutions, intolerance of different beliefs or backgrounds, and economic factors like inequality and job security. These issues have come to polarise societies, with certain regions feeling "left behind" by globalisation and thus growing hostile to it and turning to tribalism [2]. Friendshoring is a manifestation of such sentiments on a global scale, with states seeing trade as yet another realm in which to insure oneself against other nations that are seen as "them" identities.

2.1 Friendshoring on Friendly Shores

The state has always exercised serious control over economic policy around the world, especially as governments with realist inclinations often see economic security and national security as one

and the same. After all, history is strewn with examples of nations that have fallen due to economic collapse, and thus failed the theoretical imperative to survive [3]. Nations have often born great costs to force the production of goods thought to yield increasing returns or aid national security, even when considerations of comparative advantage would ostensibly suggest otherwise [4]. Friendshoring, however, noticeably differs from this rather mercantilist line of thought. It involves moving entire supply chain networks to countries regarded as geopolitically friendly, rather than trying to maximise the proportion of a country's goods produced domestically [5]. Furthermore, with friendshoring there is no illusion of its pursuit increasing efficiency. Rather, advocates argue that the cost of allowing politics to determine where goods are made is *worth the benefit* of reducing trade dependence on hostile powers.

The argument has particularly gained prominence in the context of Sino-American rivalries regarding semiconductor production and Europe's dependence on Russian oil in light of the war in Ukraine [6]. In fact, US Treasury Secretary Janet Yellen has openly described friendshoring to be a central aspect of Washington's approach to trade, stating the need to move supply lines towards "countries that have strong adherence to a set of norms and values" rather than countries with which there are geopolitical tensions. Active steps in that regard have already been taken, with trade between China and the US declining 10% between 2018 and 2020 and Apple majorly relocating its iPhone production from China to India at the behest of the American government [5, 7]. Of course, this is not to say that countries are moving towards conducting trade fully independent of any geopolitical rivals; nor is friendshoring the dominant metric by which nations make trade decisions. After all, countries do still conduct significant trade with unfriendly nations. However, much of this trade is *trade of necessity*, meaning goods for which alternative suppliers are hard to find or even unavailable [1]. Thus, it is clear that friendshoring is a phenomenon that is present, and has an observable influence on international trade that may predictably grow in the coming years as rising powers take the stage.

2.2 The Quest for Quantitative Methods in International Relations

The obvious problems of heterogeneity and endogeneity make the application of regression methods difficult in questions of international relations or political science, and when they are applied, the external validity of the results tends to be questionable. For these reasons, many would argue that assessments of foreign policy alignment and geopolitics are best left to qualitative theoretical analysis. However, when such phenomenon spill over into economic and other multidimensional domains, noise becomes ever larger, making the objectivity brought in by quantitative methods a prized quality. Furthermore, in recent years, the field has shown a growing interest in investigating phenomena such as geoeconomic fragmentation through a more econometric approach to enable working with big data and possibly serve as a unifying baseline for comparison [8].

Specifically, unsupervised machine learning with cluster analysis has served as a key tool to complement regression-based techniques, providing a more compelling account of the complex linkages in the multidimensional nexus between economics, politics, and conflict [9]. Previous literature has leveraged this to classify nations based on common characteristics using cross-sectional and dyadic data [10]. Such methods have also been applied to other fields that are accustomed to relying on qualitative analysis, with studies forming clusters using hierarchical Ward clustering and K-means clustering along with Principal Component Analysis (PCA) based on Sustainable Development Goals (SDG) indicators and specific policy indicators [11].

3 Methodology

This section will describe the data in detail, the data processing steps undertaken by this study before clustering, the implementation of the clustering analysis, and finally end by providing a methodological justification for the approach of this paper. The datasets used for this study and

their sources are listed below:

1. International Monetary Fund (IMF) International Trade in Goods by partner country (IMTS) Data: https://data.imf.org/en/datasets/IMF.STA:IMTS [12].
2. Agreement Scores Data from Erik Voeten's Dataverse in Harvard Dataverse: https://dataverse.harvard.edu/file.xhtml?fileId=11837234&version=37.0 [13, 14, 15].
3. Cultural Distance, Religious Distance, Linguistic Distance, and Geographic Distance Data from the Geo-Political Distance Data Repository, constructed by Pellegrino, Spolaore, and Wacziarg: https://www.geopoliticaldistance.org/home [16].

All the datasets listed above are dyadic by country-pairs in nature (e.g. China-India).

3.1 Data Description

3.1.1 IMF Trade Volume Data, Exports in Free on Board (FOB) US Dollars (USD)

For trade volume data, I retrieve dyadic exports data (for country dyads, excluding broad regional and economic stratifications) from 2010–2021, expressed in millions of FOB USD from the IMF's IMTS Data Explorer. Here, FOB USD indicates that the cost of goods transfers from the seller to the buyer when the cargo is loaded onto a vessel for transport [17]. The reason for choosing exports in particular is that there is a considerable lack of regular data on imports, while countries seem to report on their exports very regularly. Nevertheless, the dyadic nature of the data means that trade volume can still be calculated by summing the exports in both directions between the two countries in each dyad.

3.1.2 Agreement Scores

The Agreement Scores are essentially a spatial similarity measure of policy positions, with United Nations General Assembly (UNGA) votes being used as its proxy. Values for each dyad are calculated using Lijphart's Index of Agreement (IA) for two countries i and j, which is defined by [18]:

$$IA_{ij,t} = \frac{f_{ij,t} + \frac{1}{2} g_{ij,t}}{T_{ij,t}}$$

Where in year t,

- $T_{ij,t}$ = total votes considered with both countries
- $f_{ij,t}$ = votes in complete agreement
- $g_{ij,t}$ = votes in partial agreement (one abstains)

This yields values between 0 and 1 for all dyads.

3.1.3 Cultural Distance

The cultural distance between two countries in a dyad is defined as being the average expected disagreement between two individuals randomly drawn from those countries on a question from the World Values Survey (WVS). Firstly, we define the disagreement between two potential answers i and j as the following, accounting for binary or ordinal responses [16]:

$$\text{Disagreement}_{ij}^{q} \overset{\text{def}}{=} \begin{cases} \mathbb{I}\{i \neq j\} & \text{if } q \text{ is binary or multinomial} \\ \frac{|i-j|}{i^{q}_{\max} - i^{q}_{\min}} & \text{if } q \text{ is ordinal} \end{cases}$$

Here, i^q_{max} and i^q_{min} are the maximum and minimum values that q can take, respectively. From this, we take the expected disagreement for two countries c and c', and average it over the number of relevant questions from the WVS to yield Cultural Distance.

$$\text{Cultural Distance}_{cc'} \stackrel{\text{def}}{=} \frac{1}{Q}\sum_{q=1}^{Q}\sum_{i=1}^{i^q_{\max}}\sum_{j=1}^{i^q_{\min}} s^q_{ci}s^q_{c'j}\text{Disagreement}^q_{ij}$$

Missing values due to a mismatch in questions asked to different countries were imputed using a "Flex" method, where the authors regressed the expected disagreement of a dyad on country-pair and question fixed effects and inputted the linear predictions.

3.1.4 Religious Distance

Religious Distance was computed using the distance between two religions on a family tree of religions created by the World Religion Database. This was defined as being the length of the shortest ancestral path (LSP) from religion i to religion j divided by the sum of the LSPs between those two same religions and 0. Thus, the metric is normalised to take on values between 0 and 1. The religious distance between two countries c and c' is then defined as being the sum of the expected normalised tree distances between any two religions i and j present in those two countries, weighted by the shares of their populations adhering to those respective religions (s_{ci}, s_{cj}) [16]:

$$\begin{aligned}\text{Religious Distance}_{cc'} &\stackrel{\text{def}}{=} \sum_{i\in\mathcal{I}}\sum_{j\in\mathcal{I}} s_{ci}s_{cj}\cdot\text{Normalised Religious Tree Distance}_{ij}\\ &= \sum_{i\in\mathcal{I}}\sum_{j\in\mathcal{I}} s_{ci}s_{cj}\cdot\frac{\text{LSP}_{ij}}{\text{LSP}_{i0}+\text{LSP}_{j0}}\end{aligned}$$

3.1.5 Linguistic Distance (time-invariant)

The Linguistic Distance measure uses the family tree of languages provided by the Ethnologue, and is in fact, almost exactly symmetric to the measure for Religious Distance shown above. However, the metric for Linguistic Distance renormalises the population shares to help account for multilingualism, while the possibility of beliefs in multiple religions is naturally ruled out for Religious Distance [16]:

$$\begin{aligned}\text{Linguistic Distance}_{cc'} &\stackrel{\text{def}}{=} \sum_{i\in\mathcal{I}}\sum_{j\in\mathcal{I}}\frac{s_{ci}s_{cj}}{\sum_{i\in\mathcal{I}}\sum_{j\in\mathcal{I}}s_{ci}s_{cj}}\cdot\text{Normalised Linguistic Tree Distance}_{ij}\\ &= \sum_{i\in\mathcal{I}}\sum_{j\in\mathcal{I}}\frac{s_{ci}s_{cj}}{\sum_{i\in\mathcal{I}}\sum_{j\in\mathcal{I}}s_{ci}s_{cj}}\cdot\frac{\text{LSP}_{ij}}{\text{LSP}_{i0}+\text{LSP}_{j0}}\end{aligned}$$

Here it must be noted that unlike Religious Distance, Linguistic Distance does not vary over time, and this may prove a problem for analyses taking place over long periods, as language compositions of populations change. However, this study focuses on a rather short time frame of 6 years, meaning that this is unlikely to be an issue.

3.1.6 Geographic Distance (time-invariant)

The Geographic Distance dataset used here builds upon the CEPII GeoDist database by Mayer and Zignago, which Pellegrino, Spolaore, and Wacziarg updated with more current information. Here, the geographic distance between two countries in kilometres is defined as "the population-weighted average distance between any pairs of cities from those two countries" [16]. Several

variables were used in the dataset, from which this study used the population weighted average distance as a fraction of the earth's hemicircumference, as this normalised geographic distances in the thousands of kilometres to take values between 0 and 1.

3.2 Approach

Now that I have detailed the data sources and their nature, in this section, I will delve into the full technical approach of the study. As a general broad overview, data preprocessing steps are taken before the Ward method is applied to ensure that a dyad-level feature matrix can be built with features from all 6 datasets. This allows the application of PCA, which is necessary before clustering in order to circumvent the curse of dimensionality from such large panel data. After clustering, this study also conducts a simple Fixed-Effects First-Differences Regression as a baseline measure of the *extent* to which multidimensional alignment has affected trade in particular (friendshoring). This section will elaborate on the specifics of these steps before ending on a methodological justification to the study's approach.

3.2.1 Data Preprocessing

Here, I detail the preprocessing steps taken before conducting PCA, beginning with some simple transformations of the relevant datasets.

Converting Agreement Scores to Geopolitical Distance: While the distance parameters chosen from the four datasets constructed by Pellegrino et. al., function as intended — higher values indicating further apart dyads — we do not yet truly have a measure of geopolitical distance. The agreement scores, in fact, represent the opposite, with higher values indicating greater agreement within dyads. However, given that they take values between 0 and 1, they can be transformed into a distance parameter by simply subtracting them from 1. This can be written as $Geopolitical_{ij,t} = 1 - IA_{ij,t}$.

Calculating Trade Volume Deviations from Baseline: The trade data retrieved from the IMF contains only exports in FOB USD. Thus, the exports must be aggregated in both directions to recover bilateral trade volume: $Trade_{ij,t} = Exports_{i\to j,t} + Exports_{j\to i,t}$. Furthermore, instead of raw trade volumes in millions, I use the natural logarithm of one plus the squared deviation from the baseline differential trend established by the average trade volume for each dyad from 2010 to 2014. This ensures that the PCA does not assign an overly exaggerated weight on trade volume due to its high variance from being expressed in millions:

$$\text{Trade Volume Standardised}_{ij,t} = \ln\left[1 + \left(\text{Trade}_{ij,t} - \overline{\text{Trade}}_{ij,2010-2014}\right)^2\right]$$

Extrapolating for Missing Years: For the datasets on Cultural and Religious distances, the lack of continuous data over years proved to be an issue. The Cultural Distance dataset only contained the years 1984, 1993, 1998, 2002, 2009, 2014, and 2021 while the Religious Distance dataset contained the years 1900, 1970, 2000, 2015, 2021. However, the authors noted that to produce a panel version of their dataset, they extrapolated data for years a country wasn't surveyed by carrying over the survey data of the next closest year. Thus, I similarly carried over the data from 2014 to 2015 for Cultural Distance and from 2020 to 2021 for Religious Distance. Given that my analysis does not include *both* the extrapolated year and the source year, this should not lead to concerns of multicollinearity or lack of independence. However, it must still be acknowledged as a limit of this study and a source of inaccuracy. No such extrapolation was necessary for any other dataset.

Feature Matrix Engineering: Following data cleaning, dyads were balanced to include only country-pairs present in all six datasets for both 2015 and 2021, yielding 107 countries and 5,199 common dyads. Checks were also conducted to ensure all G20 countries were present in both years post-balancing. These dyadic distances were then transformed into country-level

feature matrices, where each country is represented as a vector containing its distances to all other countries across the six dimensions. This produced two $N \times (N \times 6)$ matrices for 2015 and 2021, with N = 107. Finally, each feature matrix was standardised to zero mean and unit variance using StandardScaler() before PCA.

3.2.2 PCA & Hierarchical Clustering (Ward)

In order to maintain consistent dimensionality reduction across *both* years of interest, the standardised feature matrices for 2015 and 2021 were first vertically stacked *before* — only then was PCA fitted to this stacked matrix. This ensured that the principal components selected were consistent between years, making clustering results for the two years comparable. Following this, the fitted PCA was applied separately to each year's standardised feature matrix.

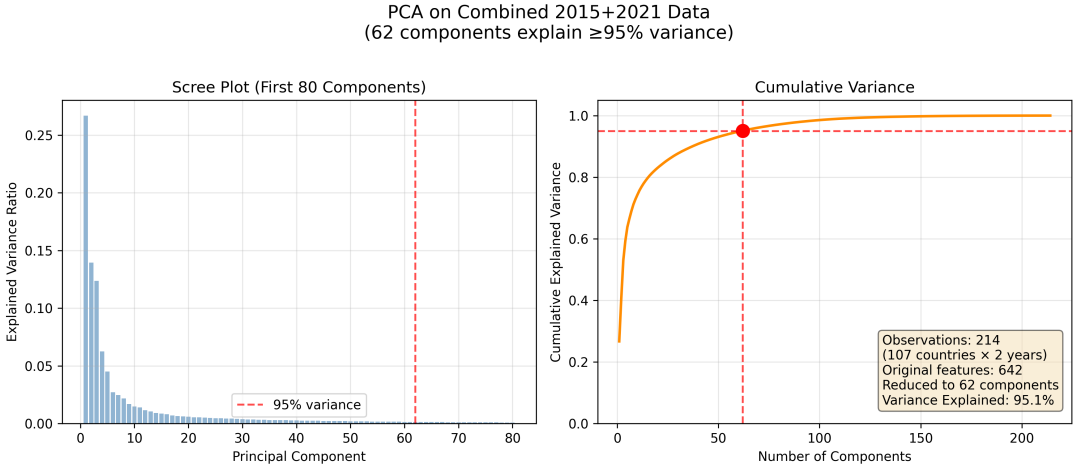


Figure 1: Scree Plot & Cumulative Variance of Fitted PCA.

As shown by the Scree Plot above, the number of components retained was determined by a cumulative explained variance threshold of 95%, which resulted in 62 principal components. This represents a significant reduction from the original $N \times 6 = 642$ features while still preserving a large portion of the variance in the dataset. Hierarchical clustering was then performed separately on the reduced PCA matrices for 2015 and 2021 using Ward linkage.

3.2.3 Evaluation Metrics

Although the literature on applying clustering to international relations seems to suggest visually slicing the dendrogram at level k based on the plausibility of the resulting groupings, this study sought to make use of a more objective metric in that regard. The number of clusters k was repeatedly evaluated over the range $k = [2, 10]$ for each year, with the optimal k determined by the level yielding the highest silhouette score [19]. For both 2015 and 2021, $k = 4$ yielded the highest silhouette scores of 0.2325 and 0.2368, respectively.

In assessing how well the clustering was able to recognise multidimensional alignment, I will conduct a qualitative comparative analysis of the produced cluster membership and real-world geopolitical groupings in 2021. This will be done through close observation of the created dendrograms and by generating world maps marked by the cluster memberships in 2015 and 2021. Comparing these maps to real-world alliances, should yield valuable insight into the state of friendshoring and geopolitical spillover — especially when complemented by regression analysis.

3.2.4 Fixed-Effects First-Differences Regression

After concluding the development of the hierarchical clustering approach, I now turn to the First-Differences Regression conducted by the study. Taking the First-Differences between t and $t-1$ gives us the following estimated model:

$$\Delta \log(\text{Trade}_{ij}) = \beta_0 + \beta_1 \Delta\text{Geopolitical}_{ij} + \beta_2 \Delta\text{Cultural}_{ij} + \beta_3 \Delta\text{Religious}_{ij} + \varepsilon'_{ij}$$

- $\Delta \log(Trade_{ij})$ is the approximate percentage change in trade volume in the dyad formed by countries i and j between 2015 & 2021
- $\Delta Geopolitical_{ij}$, $\Delta Cultural_{ij}$, and $\Delta Religious_{ij}$ all represent the changes in their respective distances between 2015 & 2021
- ε'_{ij} = error term varied by dyads

Here, it must be noted that as the regression concerns two time periods, the estimates yielded from using First-Differences or Within-Transformations are the same. Furthermore, despite the regression following a First-Differences approach, I include the constant term β_0 in the model to control for unobserved pre-trends common to all country-pairs between periods. This allows the coefficients of interest to better capture changes in trade due to these factors rather than pre-existing levels of unobservables.

Let $\Delta\mathbf{y}$ be the vector of differenced log trade flows, $\Delta\mathbf{X}$ the matrix of differenced regressors, and $\Delta\boldsymbol{\varepsilon}$ the differenced errors. I estimate the parameters under the following assumptions:

1. **Strict exogeneity:** $E[\Delta\varepsilon_{ij} \mid \Delta\mathbf{X}_{ij}] = 0$
2. **No perfect multicollinearity:** $\Delta\mathbf{X}$ has full column rank
3. **Conditional homoskedasticity:** $\text{Var}(\Delta\boldsymbol{\varepsilon} \mid \Delta\mathbf{X}) = \sigma^2\mathbf{I}$
4. **i.i.d. sampling:** $(\Delta\mathbf{X}_{ij}, \Delta\varepsilon_{ij})$ are independently and identically distributed across dyads

While the clustering analysis helps analyse whether geopolitical alliances have truly evolved into comprehensive strategic partnerships spanning multiple domains, it does not necessarily provide a clean measure of the extent to which this alignment has affected trade flows in particular. It is here that this regression can complement our analysis by serving as a simple baseline measure of the impact of friendshoring on *trade* in particular.

3.3 Methodological Justification: Warding Off Bad Clusters

When working with cross-sectional data spanning several years, it is meaningless to conduct a singular cluster analysis across all available time periods. This is because differences *across* countries during the same period will naturally be greater than differences *within* a country. That makes it likely that conducting clustering analysis on all the years of interest would yield collections of various years for the same country, yielding unhelpful conclusions, e.g. India in 2015 resembles most closely India in 2021. Wolfson et. al. notes that while the obvious solution to this is to examine every single year separately and then average across time, "it is doubtful that such a research design would provide much added value for the additional workload required" [9]. Instead, it is preferable to select individual years for separate analysis *without* averaging, as that could hide key details from systemic changes of interest. Furthermore, the reasons behind certain cluster formations may have more to do with how they behave at different time periods than how they do so in one singular period spanning several years [9]. I therefore analyse 2015

(pre–Trump baseline) and 2021 (post–COVID, pre–Ukraine war) as discrete snapshots to isolate the multidimensional alignment signal.

In selecting hyper–parameters for the clustering analysis, Ward linkage was selected based on the prevailing methodology in the relevant literature. While Ward linkage and complete linkage were both found to produce distinct clusters with large memberships, Ward was the only one that did so consistently across several years. Other linkage methods had tendencies to produce many small clusters with low memberships, which is not very meaningful as it implies there is not a lot of similarity in the data [9]. Thus, for the aforementioned reasons and as Ward linkage is considered the standard due to its minimising within-cluster variance, it was the method selected for this study.

3.4 Implementation

The code was programmed using Python 3.10.11, utilising the following libraries: `pandas 1.5.0`, `numpy 1.21.0`, `scikit-learn 1.0.0`, `statsmodels 0.14.0`, `matplotlib 3.7.0`, and `geopandas 0.13.0`. Random seed is set to 42 for reproducibility, and all necessary dependencies are listed and installable via `requirements.txt`. Furthermore, a conversion table created by Leo Picard (a PhD Candidate at the University of Basel) was used for converting Correlates of War (COW) country codes to the standard ISO 3166-1 alpha-3 codes [20].

The system is split into three main components: data preprocessing `data_loader.py`, clustering `models.py`, and evaluation through regression and the production of relevant plots `evaluation.py`. All three pipelines are called upon subsequently and in the correct order via `main.py`. Outputs produced include world maps coloured by clusters, scree plots, regression tables, and .csv files listing clusters by column for easy inspection.

4 Results

4.1 Experimental Setup

The analysis was conducted on a workstation equipped with an 11th Gen Intel Core i7-1165G7 processor, 16 GB of RAM, and an NVIDIA GeForce MX450 GPU. Python 3.10.11 served as the programming environment, with key dependencies including `scikit-learn` 1.3.2 for hierarchical clustering, `pandas` 2.1.4 for data manipulation, `statsmodels` 0.14.1 for regression analysis, and `geopandas` 0.14.3 for map visualisations. Critical hyperparameters included a random seed of 42 for reproducibility, a cumulative variance threshold of 95% for PCA component selection, and a cluster range of $k = 2$ to $k = 10$ for silhouette-based evaluation, with Ward linkage used throughout. All computations were performed locally. All raw datasets were placed in an online Zenodo repository at https://doi.org/10.5281/zenodo.18112975 for ease of access and reproducibility due to the various sources from which data was drawn.

4.2 Clustering Results, Visualisations, and Discussion

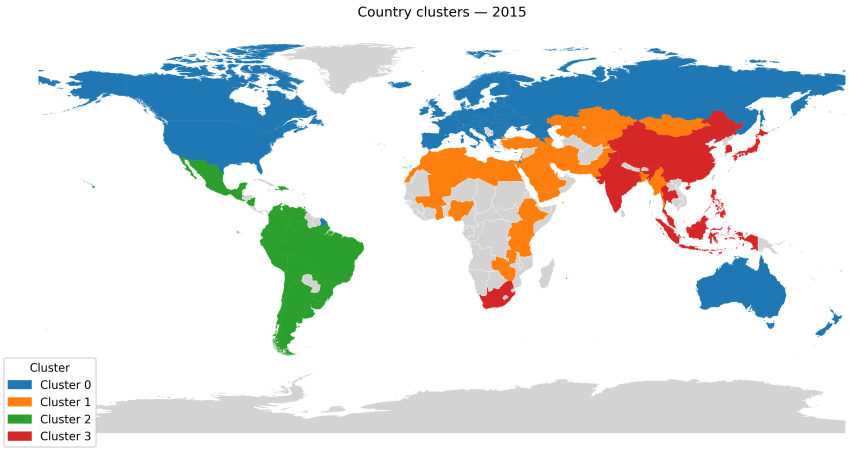


Figure 2: Cluster World Map 2015

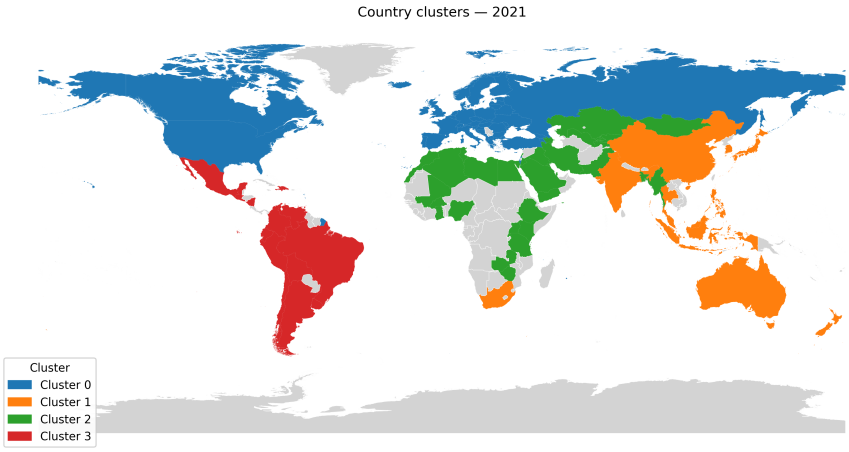


Figure 3: Cluster World Map 2021

We first start with the two world maps above, as they make it visually easier to see the clusters into which countries were grouped in our analysis. The groupings for $k = 4$ clusters are shown due to it achieving the highest silhouette score among the values tested, as noted before. As we can see, the clusters formed resemble global alliances today very little. Most glaring is the issue of Russia, Europe, and North America all being placed in the same group. This is clearly contrary to real-world alignments given Russia's invasion of Crimea and subsequent expulsion from the G8. While this may initially suggest geographic proximity to be heavily weighted in the distance metric, Australia and New Zealand was placed in the same cluster despite being significantly farther apart while Mexico was grouped together with the South American bloc despite directly bordering and being deeply entangled with the United States economically. The prospect of trade volume *itself* dominating is also a possibility, given the rather surprising placement of India, China, and Japan in the same cluster. While the three countries do trade significantly amongst one another, India and Japan are two out of four members of the QUAD — an alliance seen as being designed to counter China. However, the variance in trade volume dominating the results also seems unlikely, given that China is not placed in the same cluster as Russia despite China's disproportionately large influence on global trade flows.

Nevertheless, some subtle patterns can still be observed from these two maps. For example,

across 2015 and 2021 there most blocs largely stayed the same. Furthermore, the countries clustered together do share certain noticeable qualities. The "Western" cluster, so to speak, does include almost all of NATO (despite also including Russia in the fray). South America, as well as the Middle East, North Africa, and Sub-Saharan Africa remain as coagulated blocs across both 2015 and 2021. The Asian giants of India, China, Japan, and Korea have also all noticeably remained in the same cluster despite their geopolitical enmity. They are also joined by the unchanged membership of ASEAN in the same grouping across both years. These patterns indicate there may be some similarities between the members of the Asian bloc that do cluster them together, though perhaps not geopolitical in nature.

What is most interesting is the three cases where countries swapped clusters between the two years, namely Australia, New Zealand, and Turkey. Australia and New Zealand seemed to have moved away from a more Western-aligned bloc in 2015 towards the Pan-Asian cluster in 2021. This may however reflect growth in trade of *necessity* rather than a shift in geopolitical alignment. Australia is the source of a lion's share of Chinese imports of iron ore, while China is responsible for supplying much of Australia's rare-earth metal needs [1]. Thus, this mutual co-dependence on certain goods for which alternatives are unavailable or costly may serve as a potential explanation. Indeed, such time-varying factors or sudden upwards shifts in demand are not controlled for in the clustering analysis despite the attempt to mitigate the influence of pre-existing trade flows using the 2010-2014 baseline. Turkey's shift from the Middle Eastern cluster towards the Western bloc is also of note, given its interesting position in global politics. Although geographically being a part of the Middle Eastern World, Turkey is a nation that has exercised significant efforts towards joining with the West, as seen by its NATO membership and pursuit of European Union membership. It is possible that the clustering analysis reflects efforts to move closer towards the European/Western bloc, either as a fact of policy or as a reflection of potentially deeper macroeconomic and political shifts. However, it is not possible to draw a reliable conclusion on the matter, given the inconsistency of the clustering results when compared to real-world alliances.

Analysing the dendrograms below, however, yielded some very interesting results. Tracking the merging of the United States, starting from the lowest level here sees the US first merging with Canada, Israel, New Zealand, and Australia in that order. The next merge occurs with a cluster of strictly wealthy Western European and Scandinavian countries, thus forming a cluster that represents most of NATO. This occurs at roughly the $k = 6$ level. Similarly, turning to Russia (RUS), we see the country merging with Belarus early, which reflects real-world alliances. However, Russia seems to quickly merge with Eastern European nations like Ukraine, Poland, Latvia, and Lithuania despite its longstanding antagonistic relationship towards these countries. While it is difficult to say why this happens, it is possible that the lower cultural distances in the Eastern European region, along with strong Russo-European oil trade during this time period, has overpowered the probabilistically high geopolitical distance. Continuing at the $k = 6$ level, the South American bloc forms once more, while the previously large Asian and Middle East & African bloc splits into smaller clusters. The large Asian economies of China, India, Korea, Japan, and Singapore are grouped together, seemingly driven by their large interdependent trade flows. On the other hand, Islamic nations seem to be clustered together as well, grouping countries like Iran, Pakistan, Kuwait, Lebanon, Iraq, Qatar, Saudi Arabia, and Yemen, among others. Here, we can see the religious distance parameter at play. At the same time, poor African and Central Asian nations such as Uzbekistan, Kyrgyzstan, Mali, Rwanda, and Uganda are clustered together as well. While it is hard to suggest what factor may drive this grouping, it is also entirely possible that this cluster is driven by mutual non-membership in other clusters caused by residual grouping of low-trade/low-GDP nations.

Turning to Figure 5, which shows the dendrogram from 2021, we can start our analysis at the $k = 6$ level this time. Here, the cluster of large Asian Economies (namely China, India, Korea, Japan, and Singapore) happens very early. Furthermore, here it is made clear that Oceania's

(Australia & New Zealand) shift towards the Asian bloc happens very early as well, indicating stronger similarities than initially suspected. Another noticeable change here is that although Turkey merges with the European bloc at the higher levels, at its starting point, it merges with Israel, Russia, and Belarus — none of whom are NATO members. In fact, Turkey's eventual membership with the large Western bloc at higher k-levels comes from repeated merging with Eastern European nations in particular. While it is not possible at this stage to derive a causal explanation for this, it could be a reflection of the nation's recent authoritarian drift or decisions at the time to procure Russian-made weapons systems. Additionally, an important continuity to notice in the 2021 dendrogram is the persistence of the Latin American and Muslim blocs. Although there are minor changes in the order in which they merge between 2015 and 2021, the blocs have largely stayed stable save for the departure of Turkey and continued to form at relatively lower k-levels.

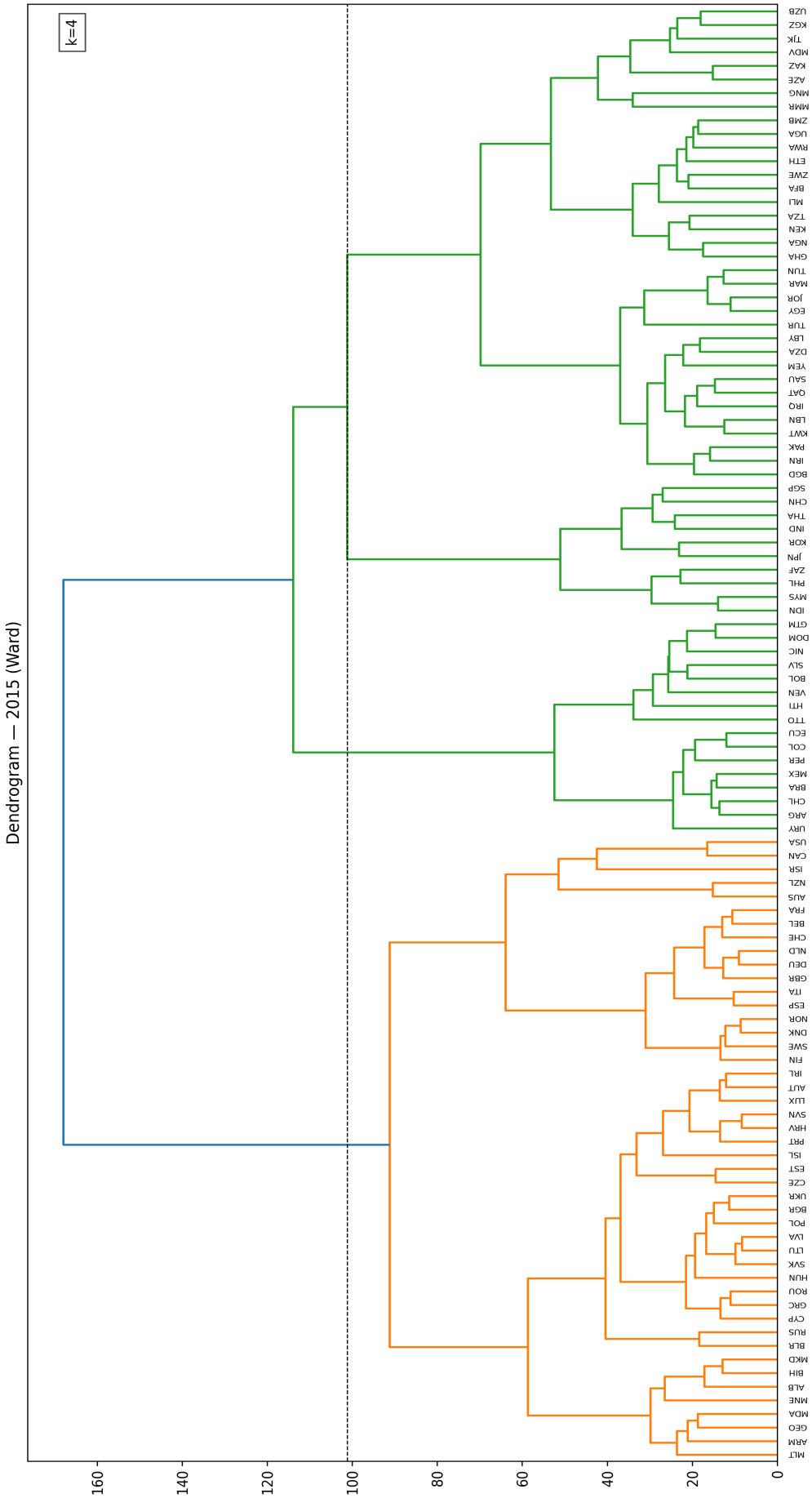


Figure 4: Hierarchical Clustering Results — 2015 Dendrogram (Horizontal)

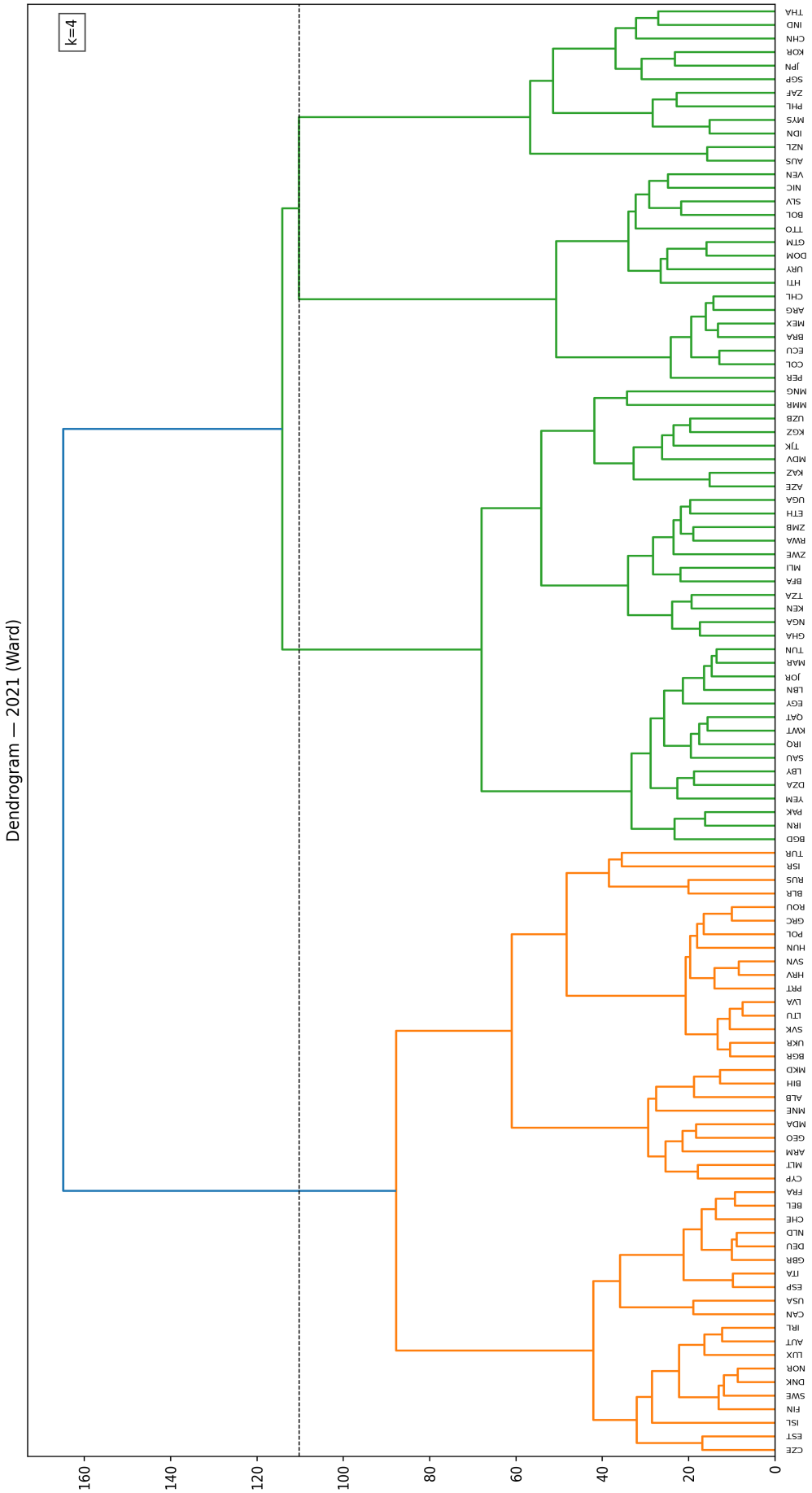


Figure 5: Hierarchical Clustering Results — 2021 Dendrogram (Horizontal)

4.3 Regression Analysis Results & Discussion

The results of the First-Differences regression conducted to complement the primary clustering analysis are displayed below in Table 1. The purpose of the regression was to add to the assessment of multidimensional alignment by helping identify not only the presence of a friendshoring effect but its extent.

Table 1: Regression Analysis Results

	Coefficient	**(S.E.)**	**t-stat**	**p-value**	**95% CI Lower**	**95% CI Upper**
Constant	0.3900***	0.0217	17.971	0.0000	0.3475	0.4325
Political Distance	-0.5605*	0.3123	-1.7948	0.0727	-1.1728	0.0517
Cultural Distance	0.0614	0.0573	1.0722	0.2837	-0.0509	0.1738
Religious Distance	-8.9644***	1.9893	-4.5062	0.0000	-12.8643	-5.0644
Observations				5199		
R^2				0.0042		
Adjusted R^2				0.0036		
F-statistic				7.2902		
F p-value				7.1e-05		

Notes: Standard errors in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Before examining the coefficients above, it is important to note that the parameters of interest here take on values between 0 and 1. Thus, it may be more helpful to interpret the coefficients in increments of 0.1. In this manner, the results infer that a 0.1 point increase in dyadic Geopolitical Distance is associated with a 5.6% decrease in bilateral trade volume, which seems reasonable and supports a friendshoring interpretation. Furthermore, the value of the constant term suggests an average of 39% trade growth per country-pair, which is plausibly close to the actual growth in global trade from 2015-2021, which was around 33.77% as per data from the World Integrated Trade Solution [21]. The Cultural Distance coefficient is shown to be statistically insignificant even at the 10% level.

However, here several caveats must be noted. Firstly, despite the high statistical significance of the coefficient for Religious Distance, it suggests that a 0.1 point increase in religious distance is correlated with an 89.6% decrease in bilateral trade volume. Such an exaggerated estimate more likely than not reflects bias, measurement errors, or a flaw in the true population model rather than true economic effect. Combined with the low R^2 value, which indicates that the estimated model explains less than 1% of the variance in trade changes, the validity of this regression analysis is likely very low.

4.4 Limitations & Future Directions

Trade flows may be largely driven by factors not captured by this model, and may instead be driven by factors such as bilateral agreements, supply-chain pragmatism, or even differences in taxation/tariff regimes. This may also explain why hierarchical clustering based on trade data did not mirror real-world geopolitical alliances or yield verifiably clear results. There are also data availability limitations that must be considered. As noted previously, the selection of years for this study was severely limited by the lack of continuity in the data. The extrapolation done for missing years in the Cultural and Religious distance datasets, along with the severe lack of import data, are just a few of the challenges faced in this regard.

Nevertheless, this project can serve as a baseline in demonstrating one potential methodology through which research in fields like international relations can make use of big data and machine learning tools. With fields like political science, sociology, and other branches of the humanities

increasingly seeking out new methods to conduct quantitative analysis, perhaps the simple pipeline established by this project can be improved and refined to support meaningful research in such areas.

5 Summary & Conclusion

This study aimed to leverage recent events to study the impact of a change in geopolitical norms and approaches on multiple dimensions, including international trade. Using a combination of unsupervised machine learning through hierarchical clustering and traditional econometric methods, this project analysed whether the growing tensions in geopolitics have had any measurable spillover effects in realms outside the direct grasp of foreign policy.

Our methodology followed the practices often used by clustering analyses in such fields of study, while also developing a data pipeline for utilising dyadic data to analyse such effects. The study's results suggest that trade-based dyadic clustering on cultural, religious, geographic, and geopolitical, and linguistic features do not successfully mirror real-world geopolitical alliances. This is shown clearly by the formed clusters often grouping together adversarial nations, such as Russia with Eastern Europe or China with India. This indicates that trade linkages more often than not survive political discord. Only three countries truly changed clusters, and even then, it is not possible to causally attribute those results to friendshoring. The weakness of the regression results further highlighted the limited explanatory power of the selected variables in accounting for changes in global trade flows.

Nonetheless, this study demonstrated a possible pathway to using machine learning tools in combination with traditional econometric techniques to explore questions at the intersection of politics, economics, and policy. The pipeline and methodology developed here provides a reproducible and sound template grounded in relevant literature for future work. Further work could experiment with average linkage methods to allow for more irregular cluster sizes, which could succeed in grouping together the numerous smaller alliances around the globe. Most importantly, while the ideal point data based on United Nations General Assembly votes has been considered the gold-standard measure of geopolitical distance in political science literature, future research could yield more direct proxies for foreign policy alignments. Such approaches could help provide a greater understanding of the linkages between economics and policy.

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